Calculus II Assignment 10

Part I. Regular Problems

1. Evaluate the iterated integral.

$$\int_0^1 \int_0^{s^2} \cos\left(s^3\right) dtds$$

2. Find the volume of the given solid: under the surface $z = xy$ and above the triangle with vertices $(1, 1), (4, 1)$, and $(1, 2)$.

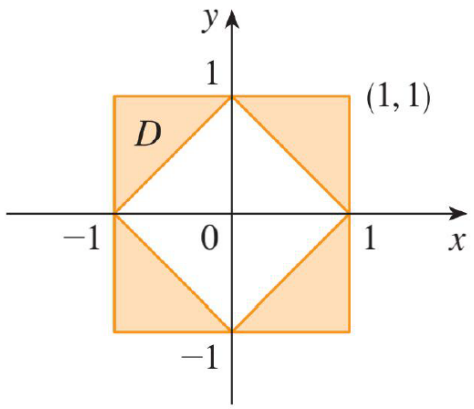
3. Sketch the region of integration and change the order of integration.

$$\int_1^2 \int_0^{\ln x} f(x, y)dydx$$

4. Evaluate the integral by reversing the order of integration.

$$\int_0^1 \int_{3y}^3 e^{x^2} dxdy$$

5. Let D be as follows.



Express D as a union of regions of type I or type II and evaluate the integral $\iint_D x^2 dA$.

6. In evaluating a double integral over a region D, a sum of iterated integrals was obtained as follows:

$$\iint_D f(x, y)dA = \int_0^1 \int_0^{2y} f(x, y)dxdy + \int_1^3 \int_0^{3-y} f(x, y)dxdy$$

Sketch the region D and express the double integral as an iterated integral with reversed order of integration.

7. Evaluate the given integral by changing to polar coordinates.

   (a) $\iint_D e^{-x^2-y^2}\,dA$, where D is the region bounded by the semicircle $x = \sqrt{4-y^2}$ and the y-axis.

   (b)
   $$\iint_R \arctan(y/x)\,dA,$$
   where $R = \{(x,y) \mid 1 \leqslant x^2+y^2 \leqslant 4, 0 \leqslant y \leqslant x\}$.

8. Use polar coordinates to find the volume of the given solid: above the cone $z = \sqrt{x^2+y^2}$ and below the sphere $x^2+y^2+z^2 = 1$.

9. Evaluate $\iint_D (x^2-y^2)\,dxdy$ where D is the square with vertices $(0,0), (1,-1)$, (1,1) and (2,0) using the change of variables $x = u+v, y = u-v$.

Part II. Group Problems

1. Evaluate the double integral.

   (a)
   $$\iint_D y\sqrt{x^2-y^2}\,dA, \quad D = \{(x,y) \mid 0 \leqslant x \leqslant 2, 0 \leqslant y \leqslant x\}$$

   (b)
   $$\iint_D x\cos y\,dA, \quad D \text{ is bounded by } y = 0, y = x^2, x = 1$$

   (c)
   $$\iint_D (2x-y)\,dA,$$
   D is bounded by the circle with center $(0,0)$ and radius 2.